

SRS RAMS KPIs

G-KPI S06001 REV.02

GBMS 2.0	COUNTRY
	ALL
DOCUMENT TYPE	LANGUAGE
KEY PERFORMANCE INDICATOR based on G-TMP A01003 rev.03	EN

Hitachi Rail – Internal

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REVISION HISTORY

Rev.	Date	Author	Approver	Authorizer	Revision Description
00	March 2022	O. Samuel Espin	R. Castagna	P. Marmo	First Issue
01	Apr 2024	O. Samuel Espin	R. Castagna	P. Marmo	Updated KPI definitions
02	Apr 2026	A. Stoica	R. Castagna	O. Hocheid	(i) migration to GBMS 2.0 (ii) new KPI definitions

DOCUMENTS TREE

L1. G-TRN S06000 SRS RAMS Transition Rules

L1. G-PRD S06000 SRS RAMS

L2. G-KPI S06001 SRS RAMS KPIs

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SCOPE

Applicability

Refer to the table below for the applicability of this document.

Legal Entities	All RSBU Legal Entities
Line of Business (LoB)	Signalling and Rail Solutions (SRS)
Project/Bid Type/Criteria	Applies to all projects (R&D, Customer, Products & Solutions Roadmap, Service etc.)
Additional Criteria and Notes	-

Effective Date

This document is applicable as soon as it is published. Each KPI is applicable as soon as the relative relevant G-PRC is published and relative relevant harmonised data sources and tools are in place.

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Technical Safety Incidents Index (TSI)

TSI-KPI-01

* Item mandatory according to ISO 22163 chapter 9.1.1.1

0. KPI Applicability

☐ Project ☐ Site ☒ Legal Entity ☐ Line of Business (LoB) ☒ Region ☒ Global

1. KPI Purpose / Improvement goal

Measures the number of Technical Safety Incidents (TSIs) identified during a defined reporting period, optionally normalized by operational activity (e.g., fleet hours, delivered units, operating time) to allow meaningful comparison over time and across products or projects. The improvement goal is to reduce the number of incidents.

Indicates the overall effectiveness of preventive and safety-related controls, safety design, and operational conditions, supporting informed decisions to strengthen process robustness and reduce the occurrence of incidents. (Lower is better.)

2. KPI definition *

Number of Technical Safety Incidents (TSIs) identified in the reporting period

Applicable for all TSIs where Hitachi has responsibility.

Technical Safety Incident (TSI), as per definitions of associated G-PRD SRS RAMS activity: **INC-01 – Manage Technical Safety Incidents.**

3. KPI calculation steps

KPI is calculated based on data extracted from TSIDB and filtered according to the KPI definition.

4. Data source & tools used for KPI consolidation and reporting

Data source: TSIDB.

Tools: TSIDB metrics and filtering functions, Excel.

5. Desired target values and defined period *

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Target: Target $\geq 10\%$ Year-over-year reduction in the safety incidents index.

Notes: Targets should be defined based on historical data, product maturity, and organizational safety objectives. Targets are internal, non-contractual, and subject to operational conditions.

Defined period: Fiscal Year

6. Responsible for report and reporting frequency *

Responsible: SRS RAMS Incident Management and Lessons Learned Authority

Reporting frequency: Monthly

7. Reporting path* and Reporting level

Reporting Path: SRS RAMS Incident Management and Lessons Learned Authority -> Head of Legal Entity -> SRS RAMS Head of Region -> SRS RAMS Head of Business Segment.

Reporting Level: Legal Entity -> Region -> Global.

8. Responsible for defining related actions *

PIRT RAMS Referent, Head of Legal Entity, SRS RAMS Head of Region, SRS RAMS Head of Business Segment.

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In-Time Containment Rate (TCR)

TSI-KPI-02

* Item mandatory according to ISO 22163 chapter 9.1.1.1

0. KPI Applicability

☐ Project ☐ Site ☒ Legal Entity ☐ Line of Business (LoB) ☒ Region ☒ Global

1. KPI Purpose / Improvement goal

Measures the percentage of Technical Safety Incidents (TSIs) for which the time between incident occurrence and containment remains within the defined maximum allowable risk exposure limit.

Indicates whether incident detection and containment actions are performed in a timely manner, enabling informed decisions on improving investigation efficiency. (Higher is better.)

2. KPI definition *

[Number of TSIs with (Actual Containment Time <= Maximum Allowable Risk Exposure Limit) ÷ Total TSIs in the reporting period] × 100

Applicable for all TSIs where Hitachi has responsibility.

Actual Containment Time: Time elapsed between *Incident Opening Date* and *Incident Containment Date*

MAXIMUM ALLOWABLE RISK EXPOSURE LIMITS	
TSI Crisis Level 3 (Severe/Catastrophic)	5 working days
TSI Crisis Level 2 (Significant/Widespread Impact)	10 working days
TSI Crisis Level 1 (Minor/Local Impact)	20 working days

with TSI Date-related attributes and Crisis Levels, as per definitions of associated G-PRD SRS RAMS activity: **INC-01 – Manage Technical Safety Incidents**.

3. KPI calculation steps

KPI is calculated based on data extracted from TSIDB and filtered according to the KPI definition.

4. Data source & tools used for KPI consolidation and reporting

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Data source: TSIDB.

Tools: TSIDB metrics and filtering functions, Excel.

5. Desired target values and defined period *

Target: Target value > 90% of TSIs within the acceptable risk exposure limit
Target range: **green** > 90%; **amber** ≤90% to 85%; **red** ≤ 85%

Notes: Targets should be defined based on historical data, product maturity, and organizational safety objectives. Targets are internal, non-contractual, and subject to operational conditions.

Defined period: Fiscal Year

6. Responsible for report and reporting frequency *

Responsible: SRS RAMS Incident Management and Lessons Learned Authority.

Reporting frequency: Monthly.

7. Reporting path* and Reporting level

Reporting Path: SRS RAMS Incident Management and Lessons Learned Authority -> Head of Legal Entity -> SRS RAMS Head of Region -> SRS RAMS Head of Business Segment.

Reporting Level: Legal Entity -> Region -> Global.

8. Responsible for defining related actions *

PIRT RAMS Referent, Head of Legal Entity, SRS RAMS Head of Region, SRS RAMS Head of Business Segment.

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Cumulative Exposure Overtime (CEO)

TSI-KPI-03

* Item mandatory according to ISO 22163 chapter 9.1.1.1

0. KPI Applicability

☐ Project ☐ Site ☒ Legal Entity ☐ Line of Business (LoB) ☒ Region ☒ Global

1. KPI Purpose / Improvement goal

Measures the cumulative amount of time that Technical Safety Incidents (TSI) remain uncontrolled beyond the maximum allowable risk exposure limit, reflecting overall timeliness of initial investigation and containment.

Indicates whether incident detection and containment actions are performed in a timely manner, enabling informed decisions on improving investigation efficiency. (Lower is better.)

2. KPI definition *

$\Sigma \max (0, \text{Actual Containment Time} - \text{Maximum Allowable Risk Exposure Limit})$

Applicable for all TSIs where Hitachi has responsibility.

Actual Containment Time: Time elapsed between TSI attributes *Incident Opening Date* and *Incident Containment Date*, as per definitions of associated G-PRD SRS RAMS activity: **INC-01 – Manage Technical Safety Incidents**.

3. KPI calculation steps

KPI is calculated based on data extracted from TSIDB and filtered according to the KPI definition.

4. Data source & tools used for KPI consolidation and reporting

Data source: TSIDB.

Tools: TSIDB metrics and filtering functions, Excel.

5. Desired target values and defined period *

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Target: Target $\geq 10\%$ Year-over-year reduction of the cumulative risk exposure overtime.

Overtime Ratio = $\Sigma \max (0, \text{Actual Containment Time} - \text{Maximum Allowable Risk Exposure Limit}) \div \Sigma \text{Maximum Allowable Risk Exposure Limit}$

Overtime Ratio ≤ 0.1 (cumulative risk exposure overtime $< 10\%$ of the overall allowable risk exposure time).

Target range: Overtime Ratio value **green** ≤ 0.1 ; **amber** ≥ 0.1 to 0.25 ; **red** > 0.25

Notes: Targets should be defined based on historical data, product maturity, and organizational safety objectives. Targets are internal, non-contractual, and subject to operational conditions.

Defined period: Fiscal Year

6. Responsible for report and reporting frequency *

Responsible: SRS RAMS Incident Management and Lessons Learned Authority

Reporting frequency: Monthly

7. Reporting path* and Reporting level

Reporting Path: SRS RAMS Incident Management and Lessons Learned Authority -> Head of Legal Entity -> SRS RAMS Head of Region -> SRS RAMS Head of Business Segment.

Reporting Level: Legal Entity -> Region -> Global.

8. Responsible for defining related actions *

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